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Deep multiple instance learning for automatic detection of diabetic retinopathy in retinal images

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Abstract: As a weakly supervised learning technique, multiple instance learning (MIL) has shown an advantage over supervised learning methods for automatic detection of diabetic retinopathy (DR): only the image-level annotation is needed to achieve both detection of DR images and DR lesions, making more graded and de-identified retinal images available for learning. However, the performance of existing studies on this technique is limited by the use of handcrafted features. The authors propose a deep MIL method for DR detection, which jointly learns features and classifiers from data and achieves a significant improvement on detecting DR images and their inside lesions. Specifically, a pre-trained convolutional neural network is adapted to achieve the patch-level DR estimation, and then global aggregation is used to make the classification of DR images. Further, the authors propose an end-to-end multi-scale scheme to better deal with the irregular DR lesions. For detection of DR images, they achieve an area under the ROC curve of 0.925 on a subset of a Kaggle dataset, and 0.960 on Messidor. For detection of DR lesions, they achieve an F1-score of 0.924 with sensitivity 0.995 and precision 0.863 on DIARETDB1 using the connected component-level validation.

1 Introduction

Diabetes now has become one of the main challenges to our human health. According to the estimation of International Diabetes Federation, the number of diabetic patients will rise up to 592 million by 2035 [1]. As a common complication of diabetes, diabetic retinopathy (DR) can cause severe vision loss and even blindness in this population. It is not only a personal catastrophe to the individual but also a threat to the nation's economy. Fortunately, timely treatment of DR can help reduce the risk of such vision loss. In order to diagnosis DR early enough for an optimal treatment, it is necessary to deploy regular DR screening in the large diabetic population. Thanks to the development of colour fundus photography, common DR lesions can be observed efficiently and non-invasively. Further, to reduce the clinical burden of examining massive retinal images, developing reliable methods for automatic detection of DR becomes an important issue in a screening programme.

Over the last two decades, numerous methods have been proposed for automatic DR detection. A comprehensive review can be found in [2]. A typical strategy adopted by most methods is to detect specific DR lesions before making a diagnose for the whole retinal image. Supervised learning techniques are usually applied to classify the segmented candidates or image patches into normal and abnormal classes. Such methods rely on the manual annotation of the DR lesions, which costs clinicians much heavier workload compared to the DR grading of retinal images. In consequence, it hinders the formation of larger training datasets, from which these data-driven methods could learn better to tackle different DR situations.

In order to save the annotation cost and make more retinal images available for learning, a weakly supervised learning technique named multiple instance learning (MIL) [3] has then been studied by different research groups [4–7]. Given only image-level labels, MIL can achieve both detection of DR images and DR lesions. Specifically, it treats retinal images as bags and their inside patches as instances. Then following the standard assumption that a DR image contains at least one DR patch and that a normal image only contains normal patches, various models have been explored to find out those discriminative DR patches [4–7]. So both classifiers for DR patches and DR images can be learned using only image-level labels. However, so far these models still use handcrafted features to characterise image patches. They fail to benefit from the larger datasets brought by MIL, often resulting in inferior detection performance compared to previous supervised learning methods.

On the other hand, deep learning techniques, especially the convolutional neural networks (CNNs), have achieved phenomenal success in natural image classification and object detection [8, 9]. By leveraging CNNs' powerful ability in representation learning, promising results are also emerging in medical image processing [10]. However, for DR detection, training CNN-based classifiers for DR lesions lacks large annotated sets of these lesions. To date, only a few studies on this route have been reported [11–13]. As an alternative, many researchers turn to achieve the automatic DR grading for retinal images, due to the availability of image-level DR grades. The whole retinal image is directly classified into a unique stage by CNNs, bypassing the detection of its inside lesions. A number of these works can be found in a Kaggle competition named 'diabetic retinopathy detection' [14], and an outstanding performance has been achieved by Graham [15]. Recently Gulshan *et al.* [16] have also demonstrated impressive results for DR detection. By training their own CNNs with an even larger and higher-quality dataset, they have achieved the state-of-the-art performance for detecting referable DR. However, CNNs trained on complete retinal images still may learn unwanted associations between DR grades and images, which can confuse clinicians when only an incorrect grade is provided for the retinal image under examination [13].

In this paper, we take the complementary advantages from deep learning and MIL, and propose a deep MIL method to achieve both detection of DR images and their inside DR lesions. Compared with existing MIL methods [4–7], our method aims at improving the detection performance by learning features from large available datasets. Compared with the whole image classification achieved by CNNs [14–16], our method provides explicit locations of DR lesions so that the detected retinal images can be easily checked by clinicians. The main pipeline of our method is described as follows. First of all, image preprocessing is applied to all retinal images, normalising factors such as image scale, illumination and

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contrast. Then image patches are extracted from the preprocessed images and fed into a CNN-based patch-level classifier to estimate their DR probabilities. After that, a global aggregation operation is applied to merge all patch-level information. It acts as an image-level classifier, generating a DR map and an image-level DR probability for each input image. At the training time, the image-level DR probability will be used to calculate the training loss. Stochastic gradient descent (SGD) and backpropagation are then used to update the CNN's parameters. Thus features and classifiers are simultaneously learned in an end-to-end way. Further, we extend our deep MIL framework from the single-scale version to a multi-scale version in order to better deal with the irregular DR lesions.

The rest of the paper is organised as follows: Section 2 presents the related work of MIL and deep MIL. Section 3 describes our proposed method in details. Section 4 presents our experiments and discussions. Section 5 concludes our method.

2 Related work

In this section, we are going to review related work in two aspects. First, we will give an overview of existing MIL methods for DR detection. Second, we will review several studies on deep MIL, especially for the medical image processing.

2.1 MIL for DR detection

In 1997, Dietterich *et al.* [3] first introduced the concept of MIL for drug activity prediction. Since then, a lot of MIL methods have been proposed. A comparative study of these methods for DR detection can be found in [7].

Xu and Li [17] first applied MIL in a content-based image retrieval (CBIR) system for DR detection. Instance prototypes were learned from the image patches to support the DR image classification. To improve the estimation for these prototypes, Venkatesan *et al.* [6] proposed a *k*NN-based non-parametric method. They also used their novel colour correlogram features [18] to make image patches more distinguishable.

In contrast, Quellec *et al.* [4] proposed a CBIR-based MIL method for DR screening. A local relevance score was computed for each image patch, measuring its similarity to the possible DR patches. Then, a global relevance score consisting of all the local ones was defined to achieve the DR image classification. Further, they developed an elaborated system for automatic detection of various retinal pathologies [5], where a diverse-density-based MIL algorithm was used as an intermediate step, trying to construct the multigranular visual words for the instance-level patterns.

Recently, two variants of the standard MIL have also been proposed for DR detection. Manivannan *et al.* [19] used a set of discriminative sub-categories to explicitly model the patches' high intra-class variations. A generalised-mean operator was applied to aggregate the patch-level DR probabilities into the image-level DR probability. Kandemir *et al.* [20] then proposed a scalable MIL method in order to handle millions of instances. For retinal images, superpixels instead of patches could be taken as the instances. However, their method only achieved the DR image classification, sacrificing the instance-level classification for its scalability.

All these methods used handcrafted features to pre-define the feature space of the instances. However, such kind of feature space may be not discriminative enough for the following MIL models [19]. In fact, even as an independent process, feature learning has proved its effectiveness on improving the MIL performance [21]. Unfortunately methods like [4, 5] still fail to exploit the benefit of feature learning, even with the access to large datasets.

2.2 Deep MIL

For supervised classification tasks, CNNs provide an efficient way to achieve the joint learning of features and classifiers [8, 9]. So it is a natural idea to extend this ability to solve weakly supervised problems. In deep MIL, CNNs are usually taken as the instance-level classifiers, and specific operations will be applied to link the bag's label to its instances' labels. Thus given only bag labels for training data, these CNNs still can be trained in an end-to-end way. For example, Oquab *et al.* [22] added a global max-pooling operation at the end of a pre-trained CNN, forming a typical deep MIL framework for weakly supervised object recognition.

For medical image processing, similar deep frameworks have also been applied to solve specific tasks. Yan *et al.* [23] used the same framework of [22] to discover discriminative patches for body part recognition. To ensure stable high responses on the discriminative patches, a spatial continuity factor was added before the global max-pooling operation. For high-resolution image classification, Hou *et al.* [24] introduced an expectation maximisation process into their deep MIL method. A bunch of discriminative patches, rather than the most discriminative ones, were selected to perform an efficient training for their CNN. Moreover, Kraus *et al.* [25] proposed a new global pooling operator, named 'Noisy-AND', in order to handle the cellular heterogeneity in microscopy image classification.

Inspired by these methods, we also attempt to use deep MIL to improve the DR detection performance. Compared to the typical method [22], our work differs in two aspects. First, instead of adapting the pre-trained CNN to solve a different task in the same natural image domain (from image classification to object recognition), we adapt such a CNN to achieve DR detection in the distinct retinal image domain. The best performance is obtained by the layer-wise fine-tuning [26] rather than the shallow tuning of [22]. Second, compared to the usual multi-scale training with random scales [22], we propose an end-to-end multi-scale scheme, which could have a more robust estimation for the DR regions and further improves the detection performance.

3 Proposed method

In this section, we will describe our deep MIL method for DR detection in details. Fig. 1 illustrates the outline of our method, which we divide into four main components: image preprocessing, CNN-based patch-level DR estimation, global aggregation for single-scale MIL and the extension for multi-scale MIL. These components are introduced in the following.

3.1 Image preprocessing

Retinal images coming from mass screening may have various image resolution, illumination and contrast. Normalising these factors helps promote the deep learning process [11]. Here we apply the image preprocessing as follows [15]. First of all, all retinal images are resized to have the same radius size of the field of view (FOV). We set this radius size $r = 384$ pixels in order to obtain an image size close to the one used in [4]. Then, a method for illumination equalisation and contrast enhancement is used:

$$I_{ic}(x, y; \sigma) = \alpha(I(x, y) - G(x, y; \sigma) * I(x, y)) + \gamma, \quad (1)$$

where a Gaussian smoothing kernel $G(x, y; \sigma)$ with standard deviation σ is applied to the resized image $I(x,y)$ in order to estimate its background illumination; factor α ($\alpha \geq 1$) is used to enhance the contrast; and an intensity γ is added to keep most pixels' intensities in the range of [0, 255]. Values of these parameters are empirically set as: $\alpha = 4$, $\sigma = r/30$ and $\gamma = 128$ [11, 15]. Each channel of the colour image I is processed with (1) independently. Finally, the image I_{ic} is cropped to a rectangular bounding box, masking out the regions within the outermost 10% of the FOV's radius, so that bright borders can be effectively removed [12]. Fig. 2 shows three examples of different cropped images before and after the illumination equalisation and contrast enhancement. These images are from the Kaggle dataset, Messidor and DIARETDB1, respectively, which will be introduced in Section 4.1.

Here we would like to note that these cropped images may have different image height, as some of them do not contain the whole retina in their original images (e.g. Figs. 2*a* and *c*). They will not be resized to have the same image height and width. Otherwise it may cause unwanted geometric distortion due to the unequal scaling along the vertical and horizontal directions.

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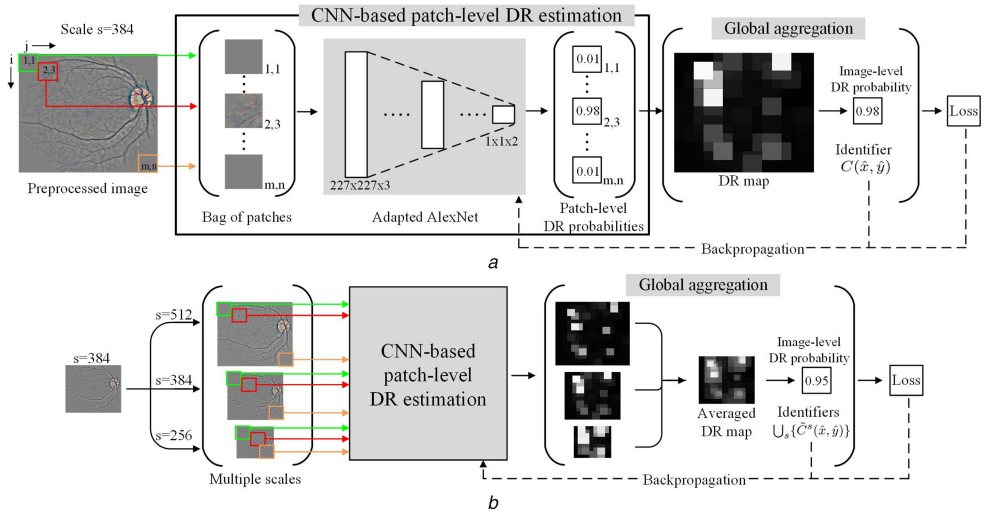


Fig. 1 *Outline of our deep MIL method for DR detection*
(a) Single-scale framework used in our first training stage, ***(b)*** End-to-end multi-scale framework used in our second training stage

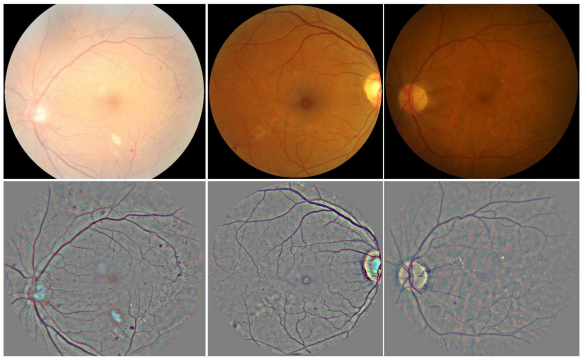


Fig. 2 *Examples of different cropped retinal images before (top row) and after (bottom row) the illumination equalisation and contrast enhancement*
(a) Image '326_left' from the Kaggle dataset, ***(b)*** Image '20051019_38557_0100_PP' from the Messidor dataset, ***(c)*** Image 'image002' from the DIARETDB1 dataset

3.2 CNN-based patch-level DR estimation

Before making a classification for the whole retinal image, we attempt to detect its DR lesions first. Image patches are regularly extracted from the preprocessed image. Then they are fed into the CNN-based patch-level classifier to estimate their DR probabilities.

Let X be a preprocessed image with a size of $H \times W$ pixels. Let the sliding window size be $K \times K$ pixels and the stride be d pixels. X is then decomposed into a bag of image patches $\{c_{i,j}\}$, where $i \in \{1, 2, \ldots, \lceil (H-K)/d \rceil + 1\}$ and $j \in \{1, 2, \ldots, \lceil (W-K)/d \rceil + 1\}$ are the vertical and horizontal index, respectively. These indexes record the patches' spatial information. For each patch $c_{i,j}$, the position of its top-left corner (h_i, w_j) is defined as follows:

$$h_i = \begin{cases} 1 + (i-1) * d & \text{if} (i-1) * d + K \le H, \\ H - (K-1) & \text{otherwise}, \end{cases} \tag{2}$$

and

$$w_j = \begin{cases} 1 + (j-1) * d & \text{if } (j-1) * d + K \le W, \\ W - (K-1) & \text{otherwise}. \end{cases} \tag{3}$$

Here each bag can have different number of patches. These patches may be resized to match the CNN's input size.

Then we adapt the pre-trained AlexNet [8] as the patch-level classifier and fine-tune it for DR detection to avoid the inefficient training from scratch [26]. In our MIL task, we have found that CNNs with random initial parameters often mis-select the normal patches of DR images for training, resulting in the vibration of parameters and a final breakdown of the training process.

The original AlexNet architecture consists of multiple functional layers, and they are well organised to achieve the feature learning and image classification. These functional layers can be roughly divided into four categories, namely the convolution layer, pooling layer, activation layer and auxiliary layer. First, the convolution layer can be a convolutional layer or a fully connected (FC) layer. Hierarchical features are learned via convolution layers. A stack of FC layers also works as a classifier which is able to approximate any function for classification. Second, the pooling layer like max pooling is applied after the convolutional layer, improving the translation invariance for learned features. Third, the activation layers include rectified linear unit (ReLU) layers and a softmax layer. Specifically, ReLU layers are added to the convolution layers (except the last FC layer), transforming the learned features non-linearly into more complex ones; and a softmax layer is added to the last FC layer, mapping its output into a probability distribution for all classes. Forth, the auxiliary layers include local response normalisation (LRN) layers and dropout layers. They are all used to regulate the responses across feature channels, improving the model's generalisation for image classification.

To accommodate our patch-level DR estimation, the pre-trained AlexNet is modified by replacing the last 1000-way FC layer with a two-way FC layer, and its filter weights are re-initialised from a Gaussian distribution with standard deviation of 0.01. For each image patch $c_{i,j}$, the terminal softmax layer will output a probability distribution $\boldsymbol{p}(c_{i,j}) = [p_1(c_{i,j})\ p_2(c_{i,j})]^{\top}$, where $p_1(c_{i,j})$ is the normal probability and $p_2(c_{i,j})$ is the DR probability. Details of this modified AlexNet architecture are shown in Table 1. Note that the dropout layer will not be added to the network unless its adjacent FC layer gets involved in our training process.

3.3 Global aggregation for single-scale MIL

After the patch-level DR estimation, we apply a global aggregation operation to combine all patch-level DR probabilities into an image-level DR map. Let M be the DR map of image X. Then in the single-scale framework (as shown in Fig. 1*a*), the value $M(x,y)$ is defined as follows:

$$M(x, y) = \max_{c_{i,j} \in G_{xy}} p_2(c_{i,j}), \tag{4}$$

where G_{xy} is the group of patches which include the pixel $X(x,y)$ in the patch extraction process:

$$G_{xy} = \{c_{u,v} | h_u \le x \le h_u + (K-1), \\ w_v \le y \le w_v + (K-1)\}.$$

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We also create an identifier map C to record the patch that contributes its DR probability to M at each location (x,y):

$$C(x,y) = \underset{c_{i,j} \in G_{xy}}{\operatorname{argmax}} p_2(c_{i,j}). \tag{5}$$

After obtaining the DR map, thresholding can be used to segment out the DR regions. Once there exists a DR region, the whole retinal image will be classified as a DR image. Therefore, for image X, its image-level DR probability $P(X)$ can be defined as the maximum value in its DR map:

$$P(X) = \max_{(x,y)} M(x,y). \tag{6}$$

For the single-scale MIL framework, it can be easily proved that $P(X)$ is equivalent to the max pooling over all patches' DR probabilities:

$$P(X) = \max_{c_{i,j}} p_2(c_{i,j}).$$

At the training time, an image label $Y \in \{1,2\}$ (where 1 is for the normal class and 2 for the DR class) will be given for each image X. Then a cross-entropy loss function is formulated as follows:

$$L = -(\mathbf{1}(Y=1)\log(1-P(X)) + \mathbf{1}(Y=2)\log P(X)), \tag{7}$$

where $\mathbf{1}(\cdot)$ is an indicator function which equals 1 if the input is true and 0 otherwise. SGD is used to minimise the loss function, and the gradients backpropagated to the CNN can be calculated as follows:

$$\frac{\partial L}{\partial P(X)} = \begin{cases} \dfrac{1}{1-P(X)} & \text{if } Y=1, \\ -\dfrac{1}{P(X)} & \text{if } Y=2, \end{cases} \tag{8}$$

and

$$\frac{\partial P(X)}{\partial \boldsymbol{p}(c_{i,j})} = \begin{pmatrix} -\dfrac{\partial P(X)}{\partial p_2(c_{i,j})} \\ \dfrac{\partial P(X)}{\partial p_2(c_{i,j})} \end{pmatrix}. \tag{9}$$

Let $(\hat{x},\hat{y})$ be the location that has the maximum value $P(X)$ of M. Then we can have:

$$\frac{\partial P(X)}{\partial p_2(c_{i,j})} = \begin{cases} 1 & \text{if } c_{i,j} = C(\hat{x},\hat{y}), \\ 0 & \text{otherwise}. \end{cases} \tag{10}$$

Therefore in the backward pass, only the patch $C(\hat{x},\hat{y})$ that contributes the maximum DR probability to the training image is involved to update the CNN's parameters, while other patches can be 'dropped out' for computational efficiency.

3.4 Extension for multi-scale MIL

DR lesions may have an irregular shape and size, and usually a multi-scale scheme is good for handling such variance. Here we propose an end-to-end multi-scale MIL framework in order to achieve a robust estimation of the DR regions and further improve the detection performance.

Instead of estimating a single DR map at the scale of the input image, we resize the input image with multiple scales and estimate their DR maps simultaneously. For the input image X, let X^s be its resized image with a specific scale $s \in \mathcal{S}$. We refer to the radius size of the image's FOV as scale s, and $\mathcal{S}$ is the set of scales used in our multi-scale framework. Let $\{c^s_{i,j}\}$ be the set of patches extracted from X^s. Therefore, image X is decomposed into a bag of multi-scale patches: $\bigcup_s \{c^s_{i,j}\}$. Following the single-scale framework, the global aggregation operation outputs a DR map M^s and an identifier map C^s for each X^s, according to (4) and (5), respectively. Then the final DR map M for image X turns to be the averaged one across all scales:

$$M(x,y) = \frac{1}{|\mathcal{S}|}\sum_s \tilde{M}^s(x,y), \tag{11}$$

where $\tilde{M}^s$ is the resized DR map of M^s, having the same image size as X. Nearest-neighbour interpolation is used, so that a corresponding resized identifier map $\tilde{C}^s$ for C^s can be easily created, keeping track of the patch that contributes its DR probability at each new location. Compared to the single-scale DR map, a more robust estimation of the DR regions can be achieved by thresholding the averaged map from multiple scales.

For our multi-scale MIL, the image-level DR probability $P(X)$ of image X is still defined as (6). However, this time it is not equivalent to the max pooling anymore. Assume that a location $(\hat{x},\hat{y})$ in M has the maximum value $P(X)$, then (6) can be reformulated as follows:

$$P(X) = \frac{1}{|\mathcal{S}|}\sum_s p_2(\tilde{C}^s(\hat{x},\hat{y})). \tag{12}$$

Let $c_{i,j}$ be an image patch which is from the bag of multi-scale patches: $c_{i,j} \in \bigcup_s \{c^s_{i,j}\}$. Then at the training time of our multi-scale MIL, the gradients backpropagated to our CNN are still formulated as (8) and (9). However, according to the relationship between $P(X)$ and $p_2(c_{i,j})$ in (12), (10) shall be changed as follows:

Table 1 Adapted AlexNet architecture for DR detection

Name	Layer types	Kernel (stride; pad)	Output
data	input	N/A	$227 \times 227 \times 3$
conv1	C + ReLU + LRN	11×11 (4; 0)	$55 \times 55 \times 96$
pool1	MP	3×3 (2; 0)	$27 \times 27 \times 96$
conv2	C + ReLU + LRN	5×5 (1; 2)	$27 \times 27 \times 256$
pool2	MP	3×3 (2; 0)	$13 \times 13 \times 256$
conv3	C + ReLU	3×3 (1; 1)	$13 \times 13 \times 384$
conv4	C + ReLU	3×3 (1; 1)	$13 \times 13 \times 384$
conv5	C + ReLU	3×3 (1; 1)	$13 \times 13 \times 256$
pool5	MP	3×3 (2; 0)	$6 \times 6 \times 256$
fc6	FC + ReLU(+dropout)	6×6 (1; 0)	1×4096
fc7	FC + ReLU(+dropout)	1×1 (1; 0)	1×4096
fc8	FC + softmax	1×1 (1; 0)	1×2

C, convolutional; ReLU, rectified linear unit; LRN, local response normalisation; MP, max pooling, FC, fully connected.

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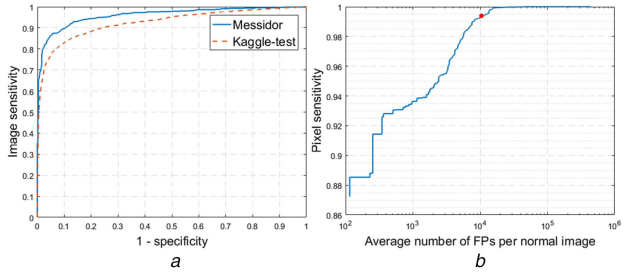


Fig. 3 *Performance plots for our deep MIL method*
(a) ROC curves on the Kaggle-test set and Messidor, ***(b)*** FROC curve on DIARETDB1

$$\frac{\partial P(X)}{\partial p_2(c_{i,j})} = \begin{cases} \dfrac{1}{|\mathcal{S}|} & \text{if } c_{i,j} \in \bigcup_s \{\tilde{C}^s(\hat{x},\hat{y})\}, \\ 0 & \text{otherwise}. \end{cases} \tag{13}$$

Therefore, for each scale the patch that overlaps the most likely DR region of the averaged DR map and that has the local maximum DR probability will be involved in the backpropagation. The gradients backpropagated to the CNN will be equally distributed to such patches from all scales. Further, in the backward pass we randomly 'drop out' at most $|\mathcal{S}|-1$ elements of $\bigcup_s\{\tilde{C}^s(\hat{x},\hat{y})\}$ for each training image, reducing the correlation between scales and improving the model's generalisation for DR detection.

4 Experimental results

4.1 Datasets

Our proposed method is evaluated on three publicly available datasets: the 'Diabetic Retinopathy Detection' dataset on Kaggle [14], the Messidor dataset [27] and the DIARETDB1 dataset [28].

The Kaggle dataset contains 35,126 graded training images and 53,576 undisclosed test images. These images are acquired under various imaging conditions. Each of them has been graded by a clinician into one of the five stages: no DR, mild, moderate, severe and proliferative DR. More details can be found in its homepage [14]. Here we use all its training images to train and evaluate our deep MIL method, and we refer to the no DR stage as the normal class and the other stages as the DR class. Following the 60-20-20 split in [11], we divide these graded images into three subsets: Kaggle-train (the first 21,076 images), Kaggle-val (the middle 7026 images) and Kaggle-test (the last 7024 images). The ratio between the number of normal and DR images in each subset is around 2.8:1, slightly imbalanced. The original FOV of each image is obtained by adaptive thresholding and circular template matching as mentioned in [11, 15].

The Messidor dataset consists of 1200 graded images whose resolution ranges from 1440×960 to 2304×1536 pixels. This dataset now is still under maintenance, and currently 547 images are graded as normal ones and 653 are graded as DR ones [27]. Here we use the whole Messidor as an independent dataset for test. Each image's FOV we use is provided by Marin *et al.* [29].

The DIARETDB1 dataset consists of 89 retinal images with a unique resolution of 1500×1152 pixels. Instead of DR grading, four experts have been asked to mark the locations of DR lesions including microaneurysms, haemorrhages, hard exudates and cotton-wool spots. Then an averaged confidence map of each kind of these lesions is provided for each image [28]. A template of the FOV is also provided in this dataset.

4.2 Experimental setup

Our deep MIL method is implemented with the MatConvNet package and its pre-trained AlexNet [30]. All experiments are conducted on a workstation with 4 NVIDIA GeForce GTX TITAN X GPUs. The Kaggle-train set is used to train our models, and the Kaggle-val set is used to monitor the training process. The area under the ROC curve (AUC) [11] is used to measure the training performance. Our method consists of two training stages. In the first training stage, we use the single-scale framework (see Fig. 1*a*) to fine-tune the pre-trained AlexNet. We set the learning rate $\lambda = 0.001$ for all its tunable layers. A mini-batch size of 32 images is applied with the momentum of 0.9, weight decay of 0.0005 and dropout rate of 0.5. This training stage is stopped when the AUC value on the Kaggle-val set no longer improves within a delay of ten epochs. In the second training stage, we select the tuned CNN with the best AUC value and further fine-tune it in our multi-scale framework (see Fig. 1*b*). Multiple scales are tried in order to find an optimal set of scales for DR detection. Only λ is reduced to 0.0001, while the other hyperparameters are kept the same as the first training stage.

At the training time, data augmentation is performed for each occurrence of the training images: random translation ([–5, 5]), scaling ([0.85,1.15]), rotation ([0°, 360°]) and horizontal flip are applied in sequence. For the patch extraction process, the side length of the sliding window is set $K = 97$ pixels and the stride $d = 49$ pixels in [4]. Besides, we oversample the DR images in the Kaggle-train set every epoch to avoid the imbalance between normal and DR classes.

In the first training stage, we achieved the best performance by deeply tuning all layers of the pre-trained CNN; and in the second training stage, we obtained the best model by further fine-tuning this deeply tuned CNN with $\mathcal{S} = \{256, 384, 512\}$ (see Section 4.5). It took us about 3 weeks to complete these two training stages. At the testing time, it takes our best model (the end-to-end multi-scale MIL with $\mathcal{S} = \{256, 384, 512\}$) 1 s to process each image.

4.3 Performance for detection of DR images

Our method's performance for DR image detection is evaluated on the independent Kaggle-test set and Messidor dataset.

All tests were made in our multi-scale framework in order to investigate the effectiveness of using multiple scales on DR detection. We switched the patch-level classifier with the best CNN learned from each training stage in order to investigate the effectiveness of using a multi-scale training strategy. We also varied the set of multiple scales $\mathcal{S}$ to investigate the effectiveness of using a multi-scale testing strategy. Further, in order to verify the effectiveness of our end-to-end multi-scale MIL, we also replace our second training stage with the usual multi-scale training [22]. We refer to this training stage as 'random-scale' MIL, as it is achieved by feeding the single-scale framework with training images resized to a random scale, which is uniformly sampled from [256, 512].

AUC and its 95% confidence intervals (95% CI) [11] are computed for each test. The pairwise AUC test mentioned in [31] is used to measure the statistical significance of the difference between these combinatorial testing strategies. The performance evaluated on the Kaggle-test set and Messidor is listed in Table 2. By fixing the learned CNN, the performance obtained by multiple test scales {256, 384, 512} is better than the single test scale {384}. By fixing the set of scales at the testing time, the performance obtained by a multi-scale training is also better than the single-scale training. Therefore using multiple scales at both the training and testing time is effective for DR detection. On either the Kaggle-test set or the Messidor dataset, we have $p < 0.01$ when comparing our end-to-end multi-scale strategy with other combinatorial strategies. Thus, our end-to-end multi-scale MIL can be considered more effective than the other strategies, including the 'random-scale' MIL. The ROC curves achieved by our proposed method on Kaggle-test and Messidor are shown in Fig. 3*a*.

As Messidor is a widely used benchmark dataset, we also compare with other methods to demonstrate the effectiveness of our deep MIL for DR detection. Results in Table 3 show that our method significantly outperforms the state-of-the-art MIL methods for DR detection, as well as some supervised methods. Further, compared to the performance of AUC = 0.844 achieved by Quellec *et al.* [5] on a large dataset, our method achieves a higher AUC = 0.925 on Kaggle-test. Besides, Gulshan *et al.* [16] achieved the state-of-the-art performance (AUC = 0.991) for referable DR detection, but they used supervised deep learning and a high-quality training set which is nearly four times larger than ours.

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Table 2 Performance on the Kaggle-test set and Messidor using single test scale and multiple test scales

Learned CNN	$\mathcal{S}$ at test	AUC on Kaggle-test	AUC on Messidor
single-scale	{384}	0.911[0.902–0.921]	0.950[0.939–0.962]
	{256,384,512}	0.913[0.904–0.922]	0.954[0.943–0.966]
our multi-scale	{384}	0.923[0.914–0.931]	0.953[0.941–0.964]
	{256,384,512}	**0.925[0.917–0.934]**	**0.960[0.950–0.971]**
random-scale	{384}	0.918[0.910–0.927]	0.953[0.941–0.964]
	{256,384,512}	0.920[0.911–0.929]	0.957[0.946–0.968]

The bold values denote the best results achieved in our comparative tests.

Table 3 Performance comparison of using different DR detection methods on Messidor

Methods	AUC on Messidor	Notes
Weakly supervised (MIL)		
our deep MIL method	0.960[0.950–0.971]	trained in Kaggle-train
Kandemir and Hamprecht [7]	0.81	two-fold cross-validation
Quellec *et al.* [5]	0.893	trained in e-ophtha
Supervised		
Roychowdhury *et al.* [5]	0.904	trained in DIARETDB1
Antal and Hajdu [32]	0.989	ten-fold cross-validation
Seoud *et al.* [33]	0.899	trained in CARA143

4.4 Performance for detection of DR lesions

DIARETDB1 is used to further evaluate our method's ability in detecting DR lesions.

First we use the connected component-level validation [12] to measure this performance. For each retinal image, its detected DR regions are divided into a set of connected components $\{D_1, D_2, \ldots, D_N\}$: $D = \bigcup_{i=1}^{N} D_i$. Its ground truth is also divided into a set of connected components $\{G_1, G_2, \ldots, G_O\}$: $G = \bigcup_{j=1}^{O} G_j$.

A pixel is considered as a true positive (TP) if, and only if, it belongs to any of the following sets:

- $D \cap G$;
- $D_i \text{ s.t. } \dfrac{|D_i \cap G|}{|D_i|} > \sigma_1$;
- $G_j \text{ s.t. } \dfrac{|G_j \cap D|}{|G_j|} > \sigma_2$.

Here we refer to $\sigma_1 \in [0, 1]$ as the compactness ratio for D_i. For a true positive set D_i, its proportion of ground-truth pixels inside should be larger than σ_1. We refer to $\sigma_2 \in [0, 1]$ as the coverage rate for G_j. For a true positive set G_i, its proportion of detected pixels inside should be larger than σ_2. Then, a pixel is considered as a false positive (FP) if, and only if, it belongs to either of the following sets:

- $D_i \text{ s.t. } D_i \cap G = \varnothing$;
- $D_i \cap \overline{G} \text{ s.t. } \dfrac{|D_i \cap G|}{|D_i|} \leq \sigma_1$.

A pixel is considered as a false negative (FN) if, and only if, it belongs to either of the following sets:

- $G_j \text{ s.t. } G_j \cap D = \varnothing$;
- $G_j \cap \overline{D} \text{ s.t. } \dfrac{|G_j \cap D|}{|G_j|} \leq \sigma_2$.

Finally, sensitivity Se, precision Pr and F1-score F1 are computed as the measurements for this performance:

$$\text{Se} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|}, \quad \text{Pr} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}|}, \quad \text{F1} = \frac{2 \cdot \text{Se} \cdot \text{Pr}}{\text{Se} + \text{Pr}}.$$

As we did not subdivide the types of detected DR lesions, here we combined the four kinds of DR lesions in DIARETDB1 together for the performance measurement. For each retinal image, a whole confidence map which represents possible lesion locations, is formed. It is achieved by taking the maximum value across the four kinds of confidence maps. Further, this whole confidence map is resized using nearest-neighbour interpolation and then cropped to match its preprocessed image. A confidence level of 0.75 is used to generate its ground truth [28], and a threshold of 0.5 is applied to its DR map to generate the DR regions. In [12], σ_1 and σ_2 were set as 0.2. However, because most microaneurysms in the ground truth have a much smaller size (around eight pixels at the scale $s = 384$) than the patches (nearly 10^4 pixels), we instead set $\sigma_1 = 0$ while $\sigma_2 = 0.9$ as a compensation. Compared to the performance of Se = 0.78, Pr = 0.78 and F1 = 0.78 achieved for exudate detection [12], our final model achieves a better performance of Se = 0.995, Pr = 0.863 and F1 = 0.924 for DR lesion detection.

We also use the free-response ROC (FROC) analysis to assess our method's performance at the pixel level. However, this time we only take the FPs on normal images into account [11]. The FROC curve is shown in Fig. 3*b*. The red dot on the curve corresponds to the performance of using a threshold of 0.5. It indicates that on average one false-positive patch would be detected on each normal image.

An example of the detected DR lesions in DIARETDB1 is shown in Fig. 4. Fig. 4*b* shows the whole confidence map with expert-marked locations of microaneurysms, haemorrhages, hard exudates and cotton-wool spots. Fig. 4*c* shows the averaged DR map estimated by our end-to-end multi-scale method. In Fig. 4*d*, the pixels inside red polygons are TPs, inside purple polygons are FPs, and inside green contours are FNs. We would like to note that these purple polygons may still contain real DR lesions which were not confirmed by the majority of the four experts. The detected DR regions may also contain neovascularisation, a kind of DR lesion that has not been annotated in DIARETDB1.

4.5 Discussions

Here we discussed several factors that may affect our method's performance for DR detection.

4.5.1 Optimal depth for fine-tuning: As demonstrated in [26], the optimal depth for CNN fine-tuning differs from one application to another. Therefore we used its layer-wise fine-tuning scheme to obtain such a CNN that could help our deep MIL method reach the best performance for DR detection. Here we searched for the optimal fine-tuning depth in the first training stage. As shown in Fig. 5, we achieved the best performance by fine-tuning all layers in our adapted AlexNet with 70 epochs. To have a fair comparison, all levels of fine-tuning were conducted 80 epochs. From Fig. 5, we can see that only tuning FC layers barely reaches a stable performance and that tuning more convolutional layers brings a consistent improvement on the detection performance. Therefore it indicates that the task difference between the image classification for natural images and the DR detection for retinal images is significant [26]. Then, in the second training stage we kept on fine-tuning all layers to obtain the final model.

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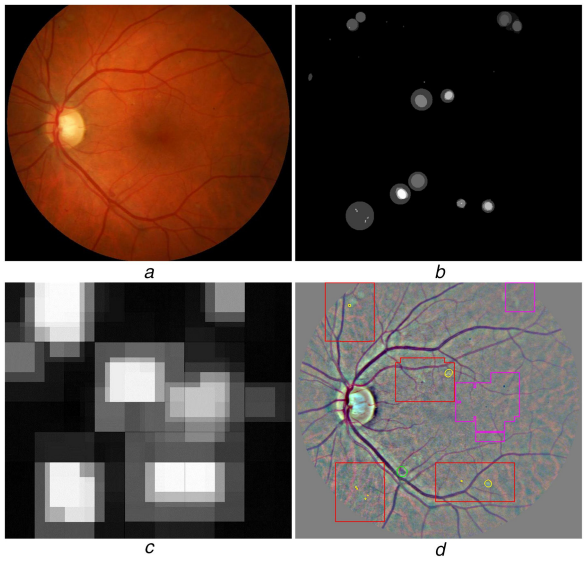


Fig. 4 *: Example of the detection of DR lesions in DIARETDB1*
(a) Image 'image054' from DIARETDB1 in the cropped bounding box, ***(b)*** Its whole confidence map, ***(c)*** Its DR map estimated by our method, ***(d)*** Detected DR regions (in red and purple polygons) and its ground truth (in yellow and green contours)

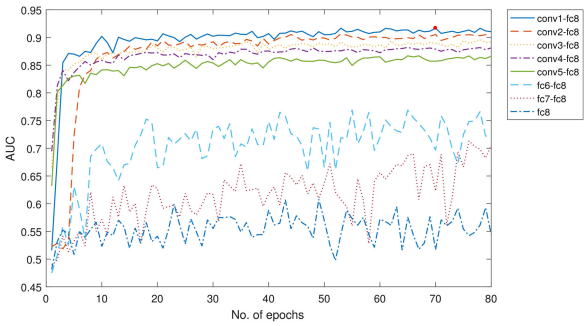


Fig. 5 *Performance of different-level fine-tuning on the Kaggle-val set during the first training stage (the red dot corresponds to the best performance)*

4.5.2 Sliding window size and stride: We also investigated the influence brought by a different setting of the sliding window size or its stride. Comparative experiments were made in our first training stage. By fixing the side length of the sliding window $K = 97$, the performance comparison of using a different stride d is shown in Fig. 6*a*. The AUC value varies slightly when $d \leq K/2$. So $d = K/2$ is a good choice for the balance between the computation cost and performance. Then by setting $d = K/2$, AUC values of using different sliding window sizes are shown in Fig. 6*b*. Currently, we achieve the best performance when K is set to be 97, a value close to the diameter of the optic disc. It is worth to note that we failed in fine-tuning our adapted AlexNet to achieve DR detection when $K = 67$. At this size, even the pre-trained AlexNet would have lots of similar responses on the normal and DR patches. So the normal patches from DR images were often wrongly selected for training, which vibrated the CNN's parameters and finally broke down the training process.

4.5.3 Combination of multiple scales: Although DR lesions vary in different sizes, proper scales are still needed in order to make DR patches distinguishable from normal patches for our deep MIL. We empirically selected several scales {192,256,384,512,640} by reference to the studies in [4, 5] and the top models used in the Kaggle competition [14, 15]. Then we tried different combinations of these scales to find out an optimal set for our deep MIL. All the experiments were conducted in our multi-scale framework. A performance comparison of using different scales is listed in Table 4. We first used the best CNN learned in our first training stage as the patch-level classifier to investigate the improvement brought by using a different set of scales at the test

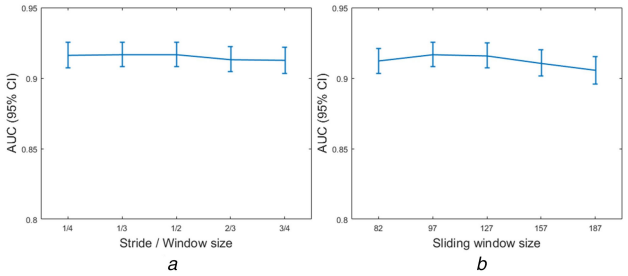


Fig. 6 *Performance comparison on the Kaggle-val set of using different sliding window sizes and strides*
(a) Performance comparison of using different sliding window strides, ***(b)*** Performance comparison of using different sliding window sizes

Table 4 Performance comparison of using different multiple scales

Learned CNN	$\mathcal{S}$ at test	AUC on Kaggle-val
Single-scale		
{384}	{384}	0.917[0.908–0.926]
{384}	{256,384,512}	0.920[0.911–0.928]
{384}	{192,256,384,512}	0.913[0.904–0.922]
{384}	{256,384,512,640}	0.919[0.910–0.927]
Our multi-scale		
{256,384,512}	{256,384,512}	**0.928[0.920–0.936]**
{192,256,384,512}	{192,256,384,512}	0.923[0.914–0.931]
{256,384,512,640}	{256,384,512,640}	0.927[0.918–0.935]

The bold values denote the best results achieved in our comparative tests.

time. From Table 4, we can see that the set {256,384,512} performs best at the test time for the CNN learned from the single-scale MIL. Adding an extra scale of 192 or 640 to this set does not bring a consistent improvement on the detection performance. On the contrary, $\mathcal{S} = \{192, 256, 384, 512\}$ causes the performance degradation. In fact, at the scale $s = 192$ the microaneurysms usually have a small size of only 1–2 pixels, so that they will be easily missed as noisy pixels, affecting the classification of DR images which only have several microaneurysms. As for $\mathcal{S} = \{256, 384, 512, 640\}$, although its performance is close to that of {256,384,512}, its computation cost doubles as it produces nearly twice as many patches as the set {256,384,512}. Then we fine-tuned the learned CNN in our multi-scale framework to further investigate whether introducing more scales at the training time improves the final performance. Results in Table 4 show that the combination {256,384,512} still performs the best. Therefore, by taking both the performance and computation cost into consideration, we prefer the set $\mathcal{S} = \{256, 384, 512\}$.

4.5.4 Feature visualisation in the CNN: To have a more concrete concept of what our CNN has learned for DR detection, we visualised the features at its convolutional layers. Fig. 7*a* shows the filters at the first convolutional layer of the pre-trained AlexNet, and Fig. 7*b* shows the corresponding filters after training in our deep MIL method. Although their shapes look the same, more weights are inclined to the red and green channels after fine-tuning, especially for the gabor-like filters. It indicates that useful information for DR detection mainly resides in the red and green channels. Figs. 7*c* and *d* show part of the features at the fifth convolutional layer, before and after training in our method, respectively. A naive image-space gradient ascent was used to visualise these features [34]. We can see that microaneurysm-like and vessel-like features are learned after training. Compared to the generic features learned from natural images, these features are more specific for DR detection. Compared to the mathematical models designed by human modelers [35], these learned features are able to handle more complex lesion situations like more than two coupled microaneurysms.

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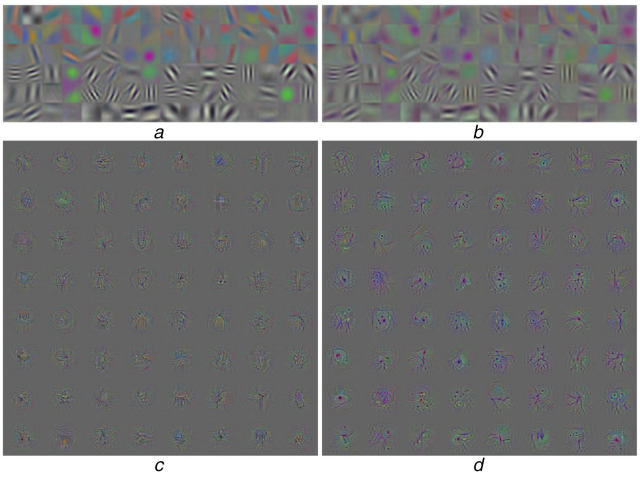


Fig. 7 *Feature visualisation in our adapted AlexNet before and after training in our method, respectively (better view in a larger size)*
(a) 96 filters at the first convolutional layer before training, ***(b)*** 96 filters at the first convolutional layer after training, ***(c)*** Features of the first 64 channels at the fifth convolutional layer before training, ***(d)*** Features of the first 64 channels at the fifth convolutional layer after training

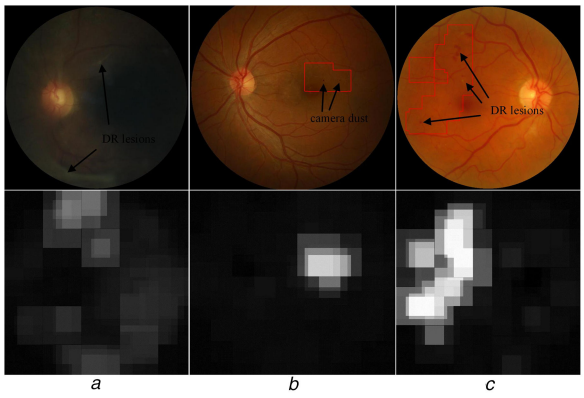


Fig. 8 *Examples of failures of our deep MIL method for DR detection (the estimated DR regions are in red polygons)*
(a) DR image '40183_left' with low illumination from Kaggle-test, ***(b)*** Normal image 'image040' with camera dust from DIARETDB1, ***(c)*** DR image '20051130_60585_0400_PP' mislabelled as normal from Messidor

4.5.5 Limitations: First, retinal images with poor image quality will be inevitably acquired in the mass DR screening. Additional image quality assessment is needed to make our method work well in such a situation. Otherwise, DR images with low illumination may be missed by our method (see Fig. 8*a*), and dark dots caused by the camera dust may be misclassified as DR lesions (see Fig. 8*b*). Second, the reliability of DR grading should be further taken into consideration [16]. Mislabelled images can cause unreliable measurement of the final performance. Fig. 8*c* shows an incorrect false-positive image in the Messidor dataset. Third, although our method is able to locate DR lesions, it is still difficult to achieve the accurate segmentation of DR lesions, even using a sliding window with a stride of 1 pixel. It restricts further quantitative analyses for the fine-grained classification of DR stages.

5 Conclusion

In this paper, we propose a deep MIL method for DR detection by taking the complementary advantages from MIL and deep learning: only the image-level annotation is needed to achieve both detection of DR images and DR lesions, meanwhile, features and classifiers are jointly learned from data. The pre-trained AlexNet is adapted and deeply fine-tuned in our framework to achieve the patch-level DR estimation. An end-to-end multi-scale framework is applied to help better handle the irregular DR lesions. Compared to existing MIL methods for DR detection, our method significantly improves the detection performance. In the future work, we are going to incorporate techniques such as semi-supervised learning and active learning into our deep MIL method to further achieve the accurate segmentation of DR lesions.

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